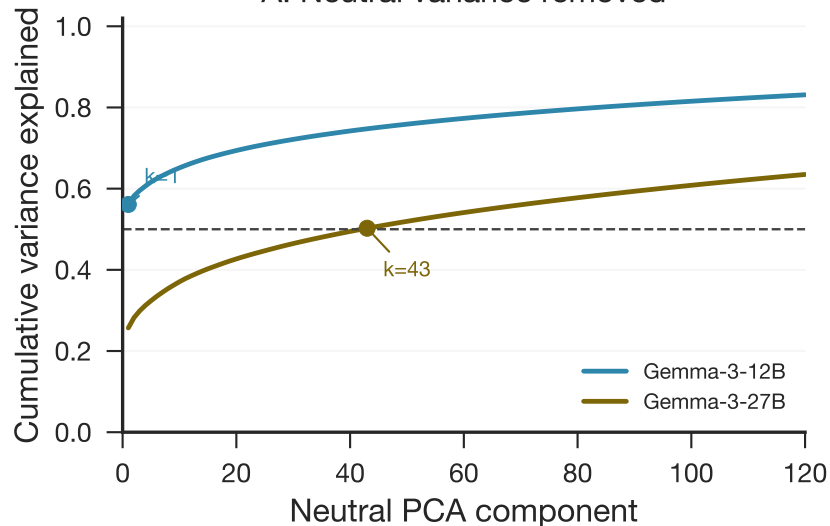
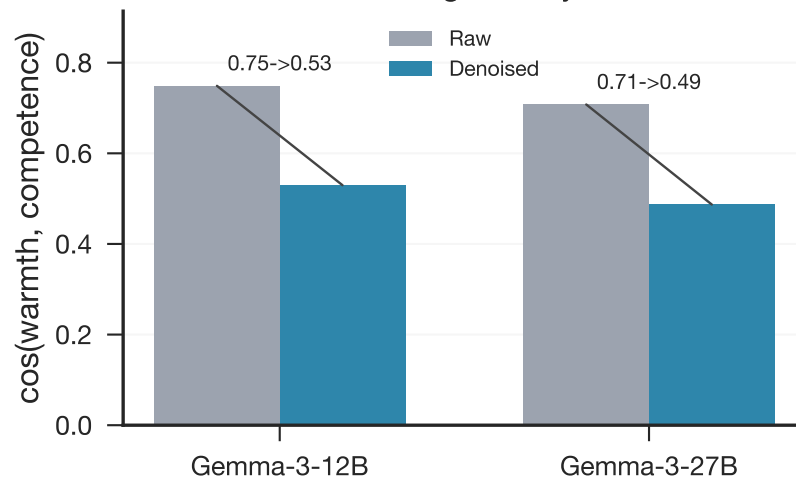


PCA denoising removes dominant neutral-variance directions from concept vectors

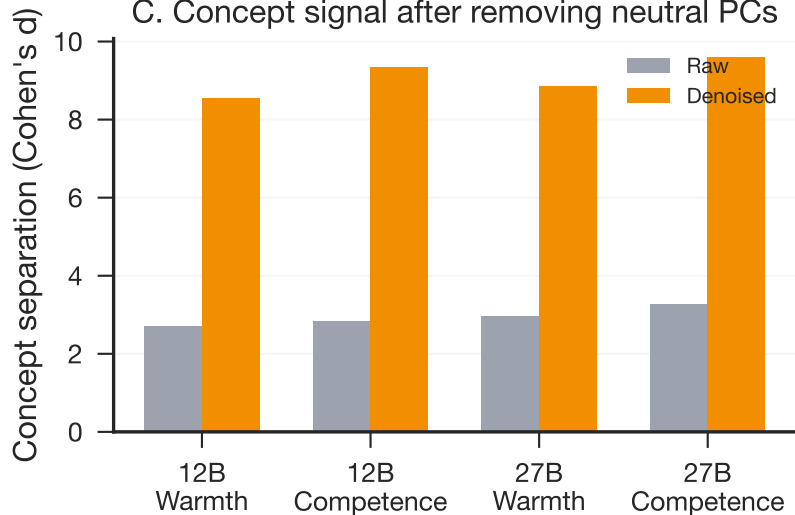
A. Neutral variance removed



B. Shared axis geometry reduced



C. Concept signal after removing neutral PCs



D. What the PCA step does

Fit PCA on neutral Wikipedia residual activations and project the top neutral-variance directions out of warmth and competence vectors.

Gemma-3-12B: 1,500 neutral texts, $d=3,840$, $k=1$ PCs (56.1% variance)
Gemma-3-27B: 1,500 neutral texts, $d=5,376$, $k=43$ PCs (50.2% variance)

Interpretation: denoising reduces shared neutral geometry, but the remaining warmth/competence cosine is not forced to zero; SCM predicts substantive correlation between the axes.